

Qualification Alignment for Postdoc Research

Focusing on Aggregate Exposure Pathway (AEP)

A component of Source to Outcome Pathway (STOP)

Raju Rimal

Outline

- ❖ My background
- ❖ Aggregate Exposure Pathway
- ❖ Data integration, analysis and visualization
- ❖ Possible contribution to the STOP application
- ❖ Publication and dissemination



My background

Education

- Bachelor
- Masters
- PhD

Experience

- Consultant
- Postdoc

Interest



Aggregate Exposure Pathway (AEP)

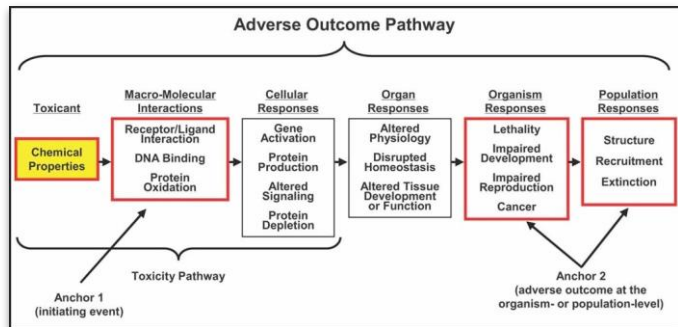
Environmental Toxicology and Chemistry

Hazard/Risk Assessment Free Access

Adverse outcome pathways: A conceptual framework to support ecotoxicology research and risk assessment

Gerald T. Ankley Richard S. Bennett, Russell J. Erickson, Dale J. Hoff, Michael W. Hornung, Rodney D. Johnson, David R. Mount, John W. Nichols, Christine L. Russom ... [See all authors](#)

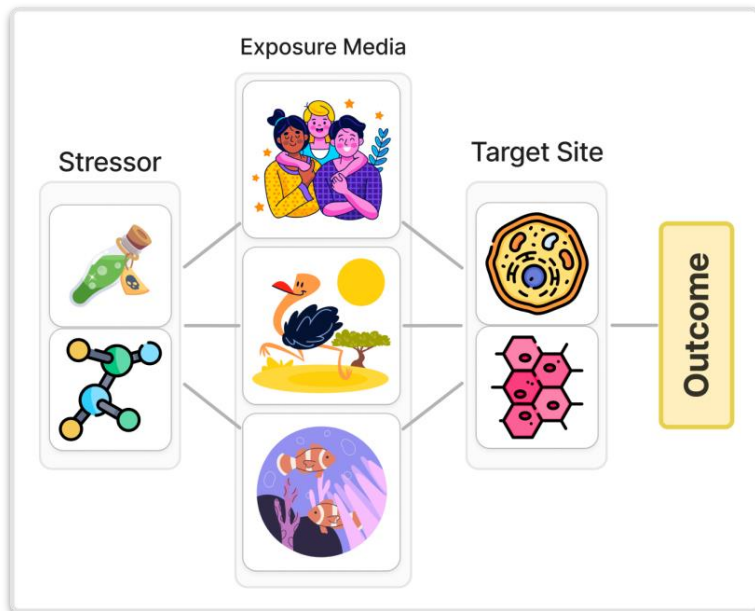
First published: 09 November 2009 | <https://doi.org/10.1002/etc.34> | Citations: 1,955



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Volume: 29
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Pages: 730-741
Published: 09.11.2009
DOI: 10.1002/etc.34



Aggregate Exposure Pathway (AEP)



Completing the Link between Exposure Science and Toxicology for Improved Environmental Health Decision Making: The Aggregate Exposure Pathway Framework

Justin G. Teeguarden^{††}, Yu-Mei Tan^{||}, Stephen W. Edwards[⊥], Jeremy A. Leonard[#], Kim A. Anderson[‡], Richard A. Corley^{††}, Molly L. Kile[§], Staci M. Simonich[‡], David Stone[‡], Robert L. Tanguay[‡], Katrina M. Waters^{††}, Stacey L. Harper^{†*}, and David E. Williams[‡]

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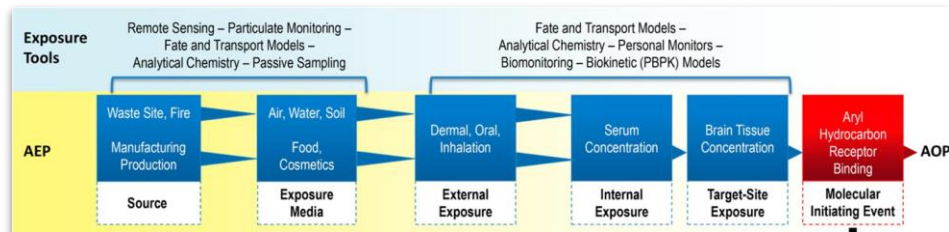
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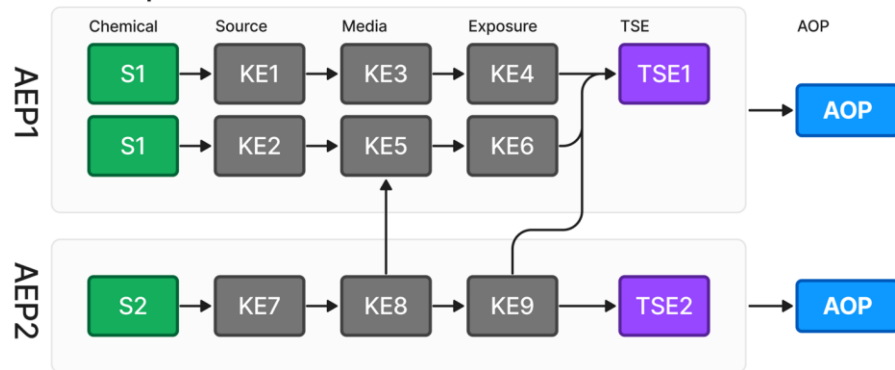
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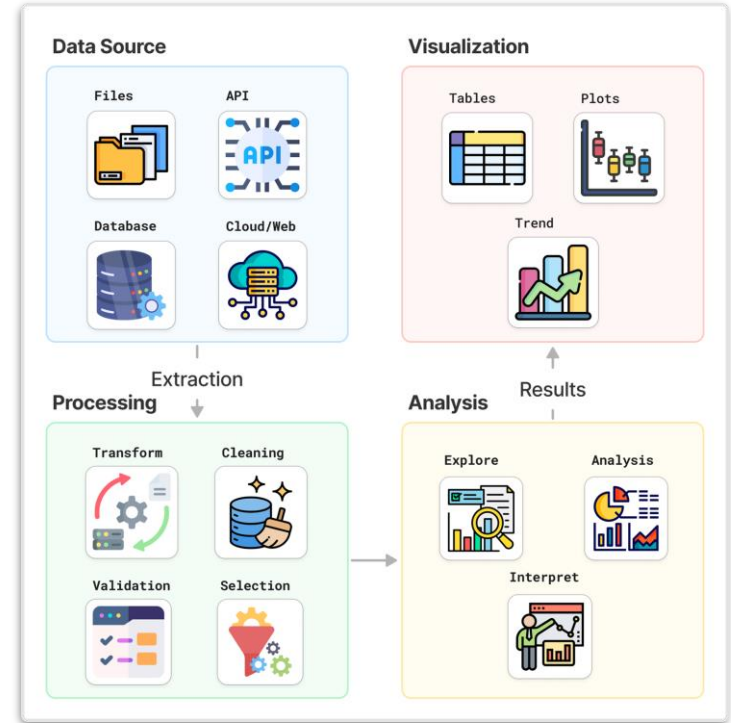
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Conceptual AEP Framework



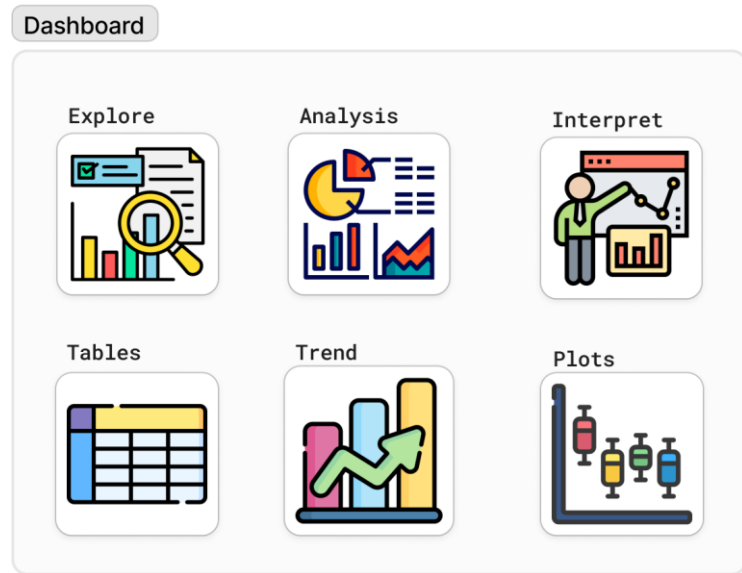
Data integration, analysis and visualization

- Many data sources
- Continuously updated data
- Data of different properties
- Integrated analysis and modelling
- Customizable visualization
- Need of data pipeline



Contributing to the STOP application

- *Improved* Graphical User Interface (GUI)
- Build *faster* Shiny App
- *Interactive* visualization of exposure, effect and risk predictions
- Informed by *better* AEP and AOP



Aligned Scientific Experiences

Research fields and publication

- Multivariate methods
- Model selection and validation
- Epidemiology
- Survival analysis

Application and package development

- Shiny application
- JavaScript based web application
- R and Python package
- Interactive plots

Thank You

NIVA (eco)toxicology postdoc interview

Raju Rimal, 2024